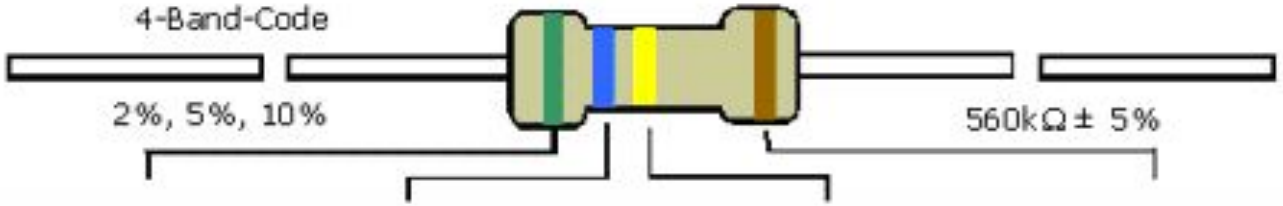
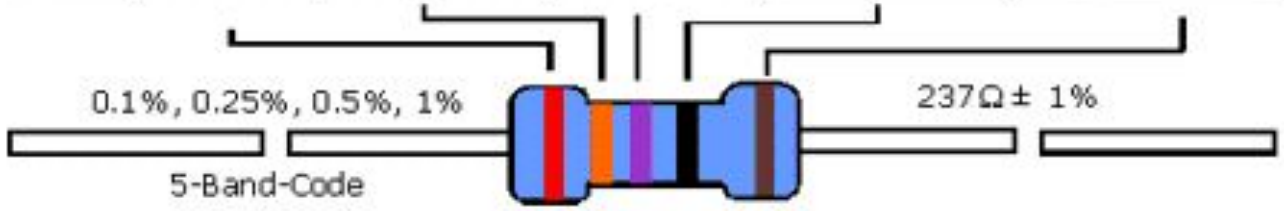
RESISTOR COLOR CODE

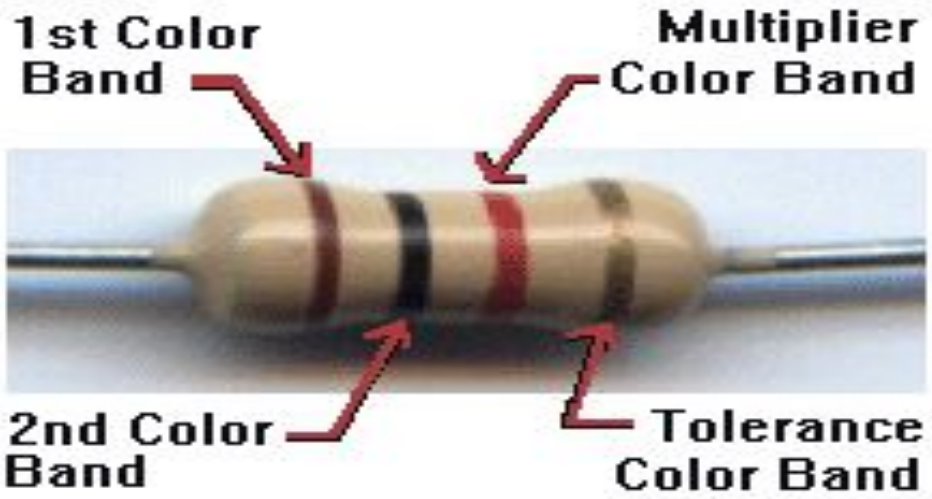
Alan L. Vergara



COLOR	1st BAND	2nd BAND	3rd BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	±0.5% (D)
Blue	6	6	6	1MΩ	±0.25% (C)
Violet	7	7	7	10MΩ	±0.10% (B)
Grey	8	8	8		±0.05%
White	9	9	9		
Gold				0.1	± 5% (J)
Silver				0.01	± 10% (K)



Resistor Color Code



1st band color gives 1st number

2nd band color gives 2nd number

3rd band color gives # of zeros

4th band color gives tolerance or ±

How to read it

- Resistors are color coded for easy reading.
- To determine the value of a given resistor look for the gold or silver tolerance band and hold the resistor with this band **to the right**.
- Then read the colored bands left to right.

How to read it

- Look at the 1st color band and determine its color. This maybe difficult on small or oddly colored resistors. Now look at the chart and match the "1st & 2nd color band" color to the "Digit it represents". **Write this number down.**

How to read it

- Now look at the 2nd color band and match that color to the same chart. **Write this number next to the 1st Digit**.
- Match the 3rd color band with the chart under multiplier. This is the number you will multiple the other 2 numbers by. **Write it next to the other 2 numbers with a multiplication sign before it.**

How to read it

- **Example:**
- First color is **red** which is 2
- Second color is **black** which is 0
- Third color is yellow which is 10,000
- Tolerance is silver which is 10%

Therefore the equation is:

2 0 x 10,000 = 200,000 Ohms ± 10%

Resistor Color Code Chart

1st. & 2nd Color Band	Digit it Represents	-----Multiplier-----
BLACK	0	x1
BROWN	1	X10
RED	2	X100
ORANGE	3	X1,000 or 1K
YELLOW	4	X10,000 or 10K
GREEN	5	X100,000 or 100K
BLUE	6	X1,000,000 or 1M
VIOLET	7	Silver is divide by 100
GRAY	8	Gold is divide by 10
WHITE	• 9	• Tolerances • Gold= 5% • Silver=10% • None=20%

How to remember the code

- Remember the color codes with this sentence: **B**rave **B**rown **R**abbits, **O**ften **Y**awn; **G**reen **B**irds
- **V**isit **G**entle oods”

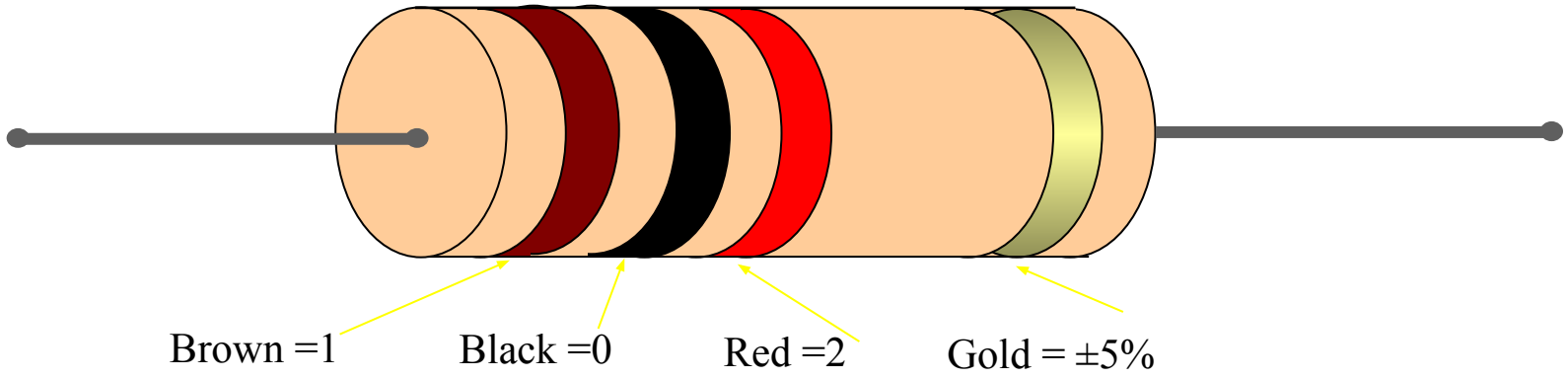
Practice

- If you have an
  - **Orange Orange Brown Gold** resistor
  - What is it's resistance?
  - The first orange means 3
  - The second orange means 3
  - The brown means 1 zero
  - The gold means ± 5%
  - **So answer is 330 ohms ± 5%**

Now you try

- Determine the resistance of each of the resistors

Example 1. Determine the nominal resistance value and the tolerance for the resistor shown below.



Solution:

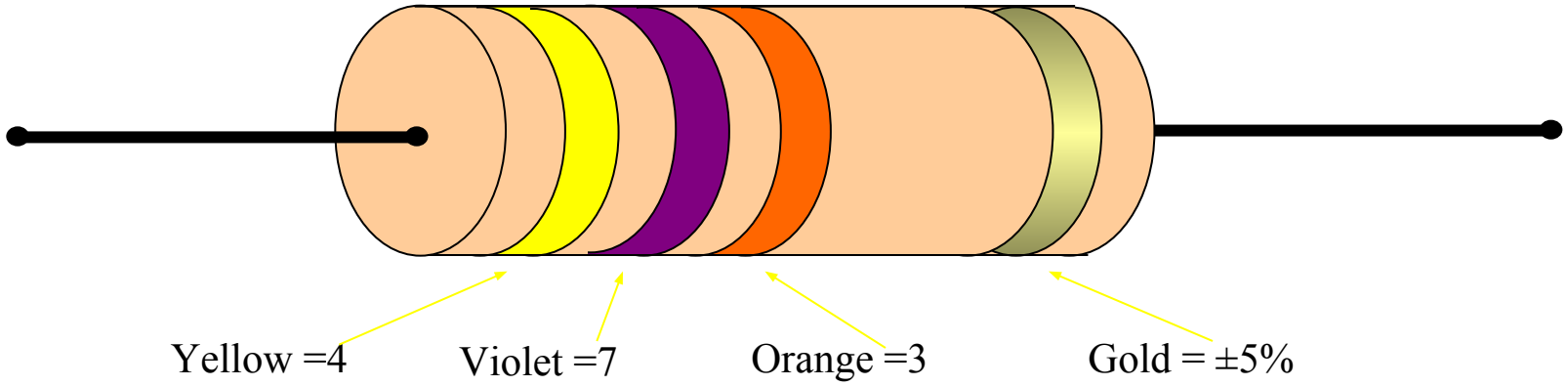
$$\underline{1}\ \underline{0} \times 10^{\underline{2}}\ \Omega \pm \underline{5}\ \%$$

Nominal value $= 10 \times 10^2 \Omega$

$= 1{,}000\Omega$

Tolerance = ±5%.

Example 2. Determine the nominal value and tolerance for the resistor below.



Solution:

$$\frac{4}{}\ \frac{7}{} \times 10^{\frac{3}{}}\ \Omega \pm \frac{5}{}\ \%$$

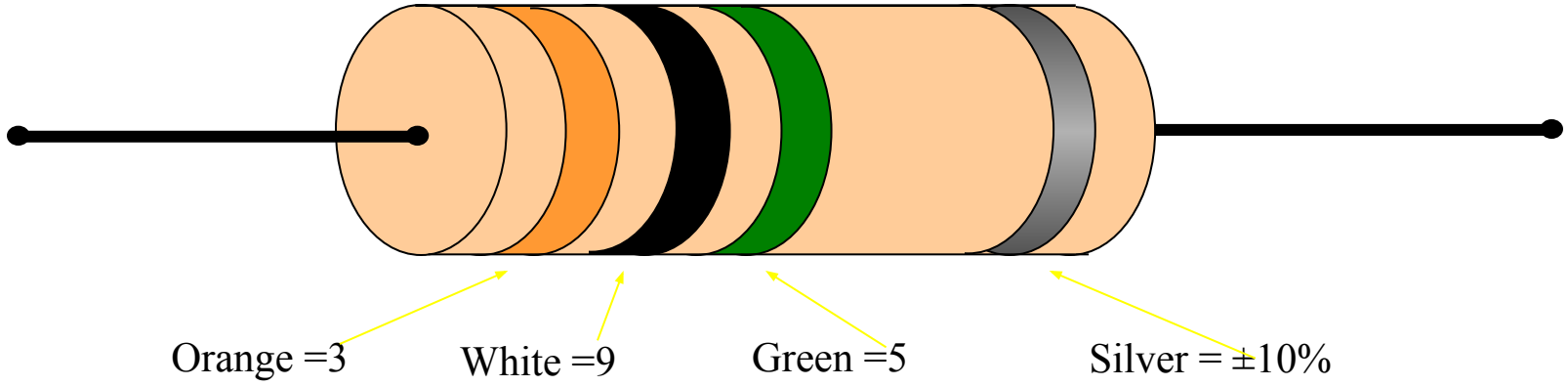
Resistor nominal value $= 47\times10^{3}\Omega$

$= 47{,}000\Omega$

$= 47\text{k}\Omega.$

Tolerance = ±5%

Example 3. Determine the nominal value and tolerance for the resistor below.



Solution:

3 × 9 10 5 Ω ± 10 %

Resistor nominal value = $39 \times 10^5 \Omega$

= 3,900,000Ω

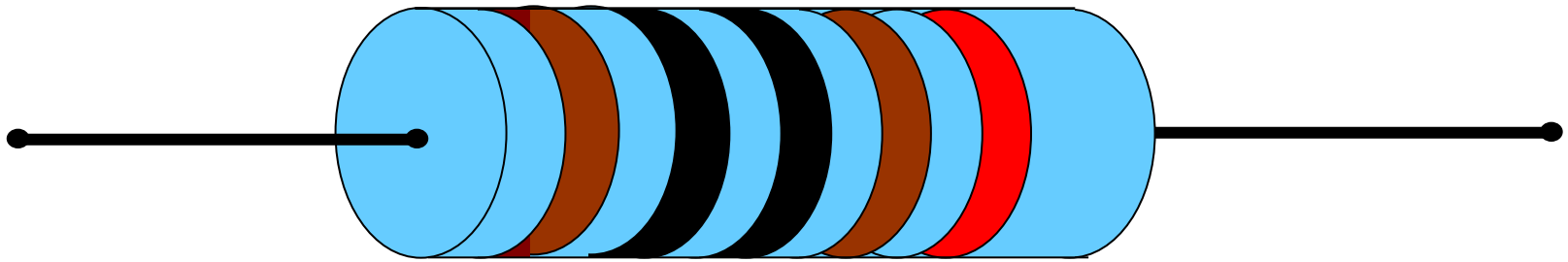
= 3.9MΩ.

Tolerance = ±10%

5-Band Resistors

- For resistors with ±1% or ±2% tolerance, the color code consists of 5 bands.

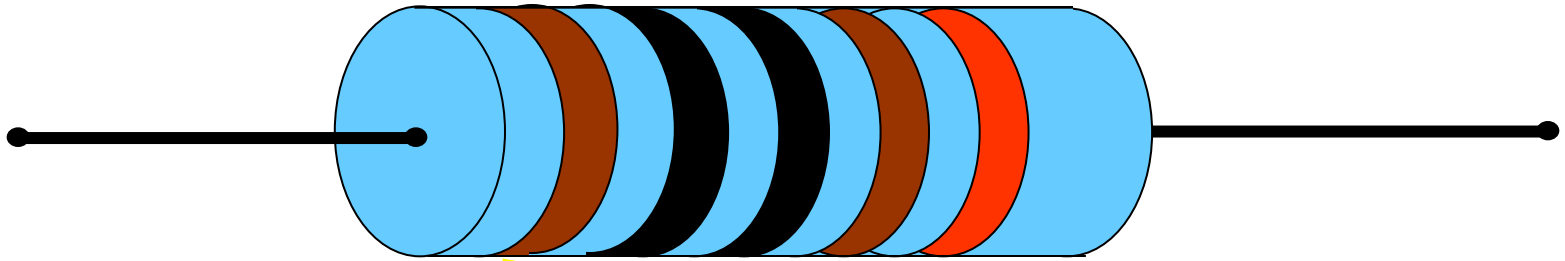
- The template for 5-band resistors is:



1st Band = 1st Significant Digit	2nd Band = 2nd Significant Digit	3rd Band = 3rd Significant Digit	4th Band = Multiplier	5th Band = Tolerance

___ ___ ___ × 10 ⎯ Ω ± ____ %

Example 4. Determine the nominal resistance and tolerance for the resistor shown below.



Solution:

$$\underline{1}\ \underline{0}\ \underline{0} \times 10^{\underline{1}}\ \Omega \pm \underline{2}\ \%$$

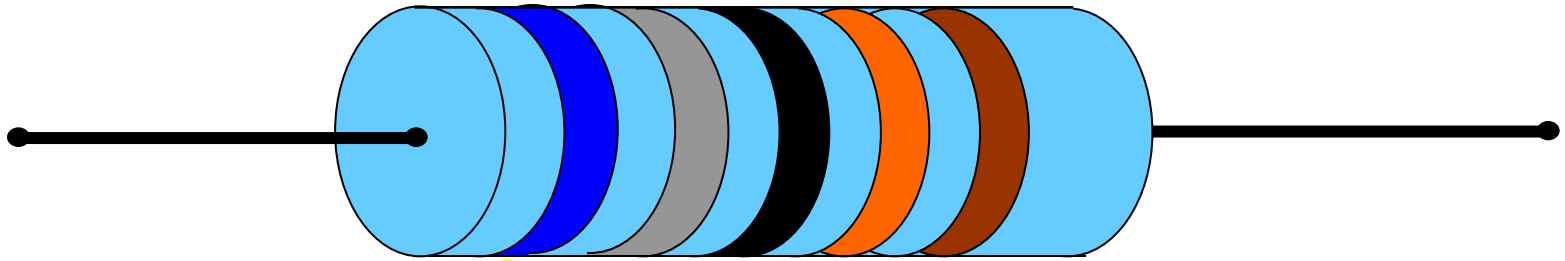
Resistor nominal value $= 100 \times 10^{1}\Omega$

$= 1{,}000\Omega$

$= 1\text{k}\Omega.$

Tolerance = ±2%

Example 5. Determine the nominal resistance and tolerance for the resistor shown below.



Solution:

$$\underline{6}\ \ \underline{8}\ \ \underline{0} \times 10^{\underline{3}}\ \Omega \pm \underline{1}\ \%$$

Resistor nominal value $= 680\times10^{3}\Omega$

$= 680{,}000\Omega$

$= 680\text{k}\Omega.$

Tolerance $= \pm 1\%$